

BUSINESS RESEARCH METHODS (BRM)

MODULE 2

Sample Design and Scale of Measurement

Sample Design

Sample design is a definite, pre-determined road map or statistical plan for selecting a subset of data (a sample) from a larger target population. It serves as the operational framework that dictates how a business will gather data, determine its sample size, and minimize errors before fieldwork begins.

The primary goal of a sample design is to ensure that the chosen sample accurately represents the broader market or population. This allows executives to make statistically sound generalizations about consumer habits, employee satisfaction, or market trends without incurring the astronomical costs and time required to survey an entire population.

Essential Elements of a Sample Design Process

Executing a robust sample design requires a business researcher to systematically navigate five key structural components:

Target Population (Universe): The total collection of all elements possessing the specific characteristics a business wants to study (e.g., all active app users in India).

Sampling Frame: The actual, accessible list or directory from which the potential respondents are pulled (e.g., a corporate employee directory or customer CRM database).

Sampling Unit: The specific, individual entity being selected for measurement, which can be a single consumer, a household, a retail outlet, or a geographic region.

Sample Size: The exact number of units to be surveyed, carefully calculated to balance statistical precision with corporate budget constraints.

Sampling Procedure: The actual statistical method or technique used to pick the individual units from the sampling frame.

Classification of Sample Design:

Sample designs are broadly categorized into two types: Non-Probability Sampling and Probability Sampling. These are further distinguished by their element selection technique, which can be unrestricted (elements drawn individually from the entire population) or restricted (all other forms).

Probability Sampling (Random)

Probability sampling, also known as random sampling or chance sampling, is a rigorous approach where every element in the population has a known, equal, or calculable chance of being included in the sample. Under this design, selection is governed by a mechanical process rather than the researcher's deliberate choice, ensuring that "blind chance" alone determines which units are picked.

Significance of the Method

Probability sampling is widely considered the superior technique for selecting a representative sample for several reasons:

Law of Statistical Regularity: It ensures that, on average, the sample will possess the same composition and characteristics as the total universe.

Elimination of Bias: Because selection is mechanical (e.g., lottery method), human bias is generally eliminated from the selection process.

Measurable Reliability: It allows researchers to mathematically measure errors of estimation and the statistical significance of the results.

Precision Control: Unlike non-probability methods, sampling error can be estimated and reduced by selecting a better design or increasing the sample size.

Suitability & Key Considerations

This method is most suitable when the research requires high accuracy and the ability to generalize findings to a larger population.

Points to consider before application:

Source List Availability: It is easiest to draw these samples when a complete list (sampling frame) of the universe is available and items are readily numbered.

Homogeneity of Population: If the population is highly variable, a simple random sample may be less precise than a stratified approach.

Budgetary Constraints: While highly reliable, these designs (especially complex ones) can be more expensive to administer than convenience-based methods.

Applicability of Specific Techniques

Probability sampling can be applied through various "mixed" or complex designs depending on the nature of the research field.

Illustration:

Illustration A: Financial Inventory Analysis (Cluster Sampling)

To estimate the proportion of defective machine parts in an inventory of 20,000 units stored in 400 cases (50 parts each), a researcher can treat the 400 cases as clusters. Instead of numbering every individual part, they randomly select n cases and examine all parts within those selected clusters. While less precise than a simple random sample, it is far more cost-effective.

Banking Efficiency Study (Multi-stage Sampling)

In a study investigating the working efficiency of nationalized banks across a country, a researcher might use a four-stage sampling plan:

Stage 1: Randomly select states.

Stage 2: Randomly select districts within those states.

Stage 3: Randomly select towns within those districts.

Stage 4: Randomly select specific bank branches within those towns.

Types of Probability Sampling

Simple Random Sampling

Meaning: Simple Random Sampling (SRS) is the most basic and widely used probability sampling method. In this technique, every element or unit in the population has an equal and independent chance of being selected into the sample.

The selection process is completely based on chance, and therefore the researcher's personal judgment or bias does not influence the selection of respondents.

In other words, each unit has the same probability of inclusion in the sample.

For example, if a population contains 500 students and the researcher wants a sample of 50 students, every student has an equal opportunity to become part of the sample.

Methods of Selection in Simple Random Sampling

1. Lottery Method

Under this method:

Names or numbers of all population units are written on slips.

The slips are mixed thoroughly in a box or container.

Required number of slips are picked randomly.

This method is suitable for small populations.

Example

A teacher writes the names of 100 students on slips and randomly picks 10 slips for participation in a competition survey.

2. Random Number Table Method

In this method:

Population units are assigned numbers.

Random number tables are used to select the sample.

This is more scientific and suitable for larger populations.

3. Computer-Generated Random Numbers

Modern researchers commonly use:

Excel

SPSS

R software

Python

Online random generators

These tools automatically generate random samples quickly and accurately.

Features of Simple Random Sampling

1. Equal Chance of Selection

Every unit has identical probability of selection.

Importance

This ensures fairness and representativeness.

2. Freedom from Personal Bias

Selection depends entirely on randomization and not on researcher preference.

Importance

This improves objectivity and reliability.

3. Simplicity

The method is easy to understand and implement.

Importance

Suitable for beginners and basic research studies.

4. Representative Nature

If properly conducted, the sample adequately represents the population.

5. Basis for Statistical Analysis

Most statistical tools and tests assume random sampling.

Therefore, SRS supports:

Hypothesis testing

Confidence intervals

Regression analysis

Correlation analysis

Applicability of Simple Random Sampling

Simple random sampling is appropriate under the following conditions:

1. Homogeneous Population

When population members are similar in characteristics, SRS gives effective representation.

Example

Employees working in the same department with similar qualifications.

2. Availability of Complete Sampling Frame

The researcher must possess a complete list of all population elements.

Example

Student register

Employee database

Electoral roll

Without a complete list, random selection becomes difficult.

3. Small or Moderate Population Size

SRS is more practical when the population size is manageable.

4. Quantitative Research Studies

Widely used in:

Social science research

Educational studies

Market research

Opinion surveys

Illustration

Suppose a college has 500 students and a researcher wants to study students' digital learning preferences.

Steps followed:

Obtain the complete list of 500 students.

Assign numbers from 1 to 500.

Use a lottery method or computer-generated random numbers.

Select 50 students randomly.

Thus, every student gets an equal opportunity to be selected.

Advantages of Simple Random Sampling

1. Easy to Understand and Use

The procedure is straightforward and uncomplicated.

2. Free from Selection Bias

Since selection is random, chances of favouritism are minimized.

3. Scientifically Reliable

Produces dependable and valid results if sample size is adequate.

4. Suitable for Statistical Analysis

Sampling error can be estimated mathematically.

5. Representative Sample

When population is homogeneous, the sample reflects population characteristics effectively.

Disadvantages of Simple Random Sampling

1. Requires Complete Sampling Frame

The researcher must have a full list of population units.

Problem

In many real-life situations, obtaining such a list is difficult.

2. Not Suitable for Large Scattered Populations

If population units are geographically dispersed, random selection becomes costly and time-consuming.

Example

Selecting farmers from all villages across India.

3. May Not Represent All Subgroups

Important categories may be accidentally underrepresented.

Example

A random sample may include fewer female respondents unintentionally.

4. Time Consuming for Large Populations

Preparing lists and conducting randomization may require substantial effort.

Conclusion

Simple Random Sampling is the foundation of probability sampling techniques. It is highly scientific, objective, and reliable because every population unit has an equal chance of selection. The method works best when the population is homogeneous and a complete sampling frame is available. Although it has limitations in large and geographically dispersed populations, it remains one of the most widely used sampling methods in academic and business research.

Systematic Sampling:

B. Systematic Sampling

Meaning

Systematic sampling is a probability sampling technique in which the researcher selects every k th element from the population after choosing a random starting point.

Instead of selecting units completely randomly, the sample is selected at regular intervals.

The sampling interval is calculated as:

$$k = N/n$$

Where:

(N) = Total population size

(n) = Required sample size

(k) = Sampling interval

Procedure of Systematic Sampling

Prepare a complete list of population elements.

Determine the sample size.

Calculate the sampling interval.

Select a random starting point.

Choose every k th unit thereafter.

Example

Suppose a company has 1,000 employees and the researcher needs a sample of 100 employees.

Sampling interval:

$$K = 1000/100 = 10$$

If the random starting point is 4, then the selected respondents will be:

4, 14, 24, 34, 44, and so on.

Features of Systematic Sampling

1. Regular Interval Selection

Units are selected systematically after fixed intervals.

2. Simpler Than Random Sampling

Less complicated than selecting every unit through separate randomization.

3. Uniform Coverage

Ensures sample is spread throughout the population.

4. Probability-Based Method

Every unit still has a known chance of selection.

Applicability

Suitable when:

Population list is available

Population is arranged systematically

Quick selection is required

Common Areas

Employee surveys

Production quality inspection

Customer database studies

Advantages

Simple and fast

Economical

Easy to administer

Better population coverage

Disadvantages

Risk of periodicity bias

Not suitable if hidden patterns exist in the list

Example

If every 10th employee belongs to the same department, results may become biased.

Conclusion

Systematic sampling is an efficient and practical sampling method that simplifies sample selection while maintaining randomness. It is especially useful in organized populations where lists are available.

Stratified Sampling

Meaning

Stratified sampling is a probability sampling method in which the population is divided into homogeneous subgroups called strata, and samples are selected separately from each stratum.

Each stratum represents a particular category or characteristic of the population.

Purpose of Stratification

The main objective is to ensure that all important groups in the population receive proper representation in the sample.

Procedure of Stratified Sampling

Divide the population into strata.

Ensure each stratum is homogeneous internally.

Select samples from each stratum randomly.

Combine selected units to form the final sample.

Example

A university has:

700 undergraduate students

300 postgraduate students

The researcher selects samples separately from both groups.

Features of Stratified Sampling

1. Division into Strata

Population is classified into meaningful categories.

2. Homogeneity Within Strata

Members inside each stratum are similar.

3. Representation of All Groups

Every subgroup receives representation.

4. Greater Accuracy

Produces more precise estimates than simple random sampling.

Types of Stratified Sampling

1. Proportionate Stratified Sampling

Sample size from each stratum is proportional to population size.

2. Disproportionate Stratified Sampling

Equal or unequal numbers are selected irrespective of population proportion.

Applicability

Suitable when:

Population is heterogeneous

Subgroups differ significantly

Representation of all categories is important

Common Areas

Educational research

Income group studies

Gender-based surveys

Market segmentation studies

Advantages

Better representation

Higher precision

Reduced sampling error

Useful for subgroup analysis

Disadvantages

Requires detailed population information

Classification process may be difficult

More complex administration

Conclusion

Stratified sampling is one of the most accurate probability sampling methods because it ensures representation from all major subgroups. It is highly suitable for heterogeneous populations.

Cluster Sampling

Meaning

Cluster sampling is a probability sampling method in which the population is divided into clusters, and some clusters are selected randomly for study.

A cluster is a naturally occurring group of population elements.

Instead of selecting individual units directly, entire groups are selected.

Nature of Clusters

Clusters should:

Represent the population adequately

Be heterogeneous internally

Be similar to one another

Procedure of Cluster Sampling

Divide population into clusters.

Randomly select clusters.

Study all units or selected units within chosen clusters.

Example

A researcher studying rural healthcare may:

Divide villages into clusters

Randomly select 20 villages

Survey households within those villages

Features of Cluster Sampling

1. Group-Based Selection

Entire groups are selected instead of individual units.

2. Economical

Reduces travel and administrative expenses.

3. Suitable for Geographically Scattered Populations

Useful when population is spread over wide areas.

Applicability

Used in:

National surveys

Rural development studies

Educational surveys

Public health research

Advantages

Cost-effective

Saves time

Operationally convenient

Suitable for large populations

Disadvantages

Less accurate than stratified sampling

Higher sampling error possible

Clusters may not perfectly represent population

Conclusion

Cluster sampling is a practical method for large-scale and geographically dispersed populations. Although it may sacrifice some accuracy, it significantly reduces operational cost and complexity.

Multistage Sampling

Meaning

Multistage sampling is an advanced probability sampling method in which sampling is carried out in multiple stages using smaller and smaller sampling units at each stage.

Instead of selecting final respondents directly, the researcher selects samples step by step.

Procedure of Multistage Sampling

Selection occurs in stages.

Example

For a national education survey:

Select states

Select districts within states

Select schools within districts

Select students within schools

Features of Multistage Sampling

1. Sampling in Several Stages

Selection occurs progressively.

2. Combination of Methods

Different sampling methods may be used at different stages.

3. Flexible Design

Can be adapted according to research needs.

4. Suitable for Large Populations

Useful for nationwide or large-scale studies.

Applicability

Used in:

Census surveys

Government studies

National health surveys

Educational research

Large market research projects

Advantages

Practical and flexible

Reduces cost and effort

Suitable for very large populations

Administrative convenience

Disadvantages

Complex procedure

Higher possibility of sampling error

Requires careful planning

Difference Between Stratified, Cluster, and Multistage Sampling

Overall Conclusion

Systematic, stratified, cluster, and multistage sampling are important probability sampling techniques widely used in research and statistical investigations.

Systematic sampling provides simplicity and uniformity.

Stratified sampling improves representativeness and accuracy.

Cluster sampling reduces operational cost in scattered populations.

Multistage sampling offers flexibility for large-scale studies.

The selection of an appropriate sampling method depends on:

Research objectives

Population structure

Availability of resources

Required accuracy and precision.

2. Non-Probability Sampling (Non-Random)

A non-probability sampling design is a sampling procedure that does not afford any mathematical basis for estimating the probability that each item in the population has of being included in the sample. It is also widely referred to as deliberate sampling, purposive sampling, or judgement sampling.

Unlike probability sampling—which relies on blind chance and mechanical selection—non-probability sampling places the researcher's choice supreme. The organizers of the inquiry purposefully select specific units from the universe based on the assumption that the chosen small mass will be typical or representative of the whole population.

Characteristics:

Supreme Researcher Discretion: The selection of individual units is driven by deliberate choice rather than a randomized mechanism.

High Vulnerability to Bias: Because selection depends on human judgment, the personal element has a strong chance of entering the process. If an investigator selects a sample to yield results favourable to a specific viewpoint, the entire inquiry can become vitiated.

Non-Measurable Sampling Error: In this design, sampling error cannot be mathematically estimated or calculated.

No Formal Statistical Inferability: The resulting samples do not possess the characteristics of random samples. Consequently, any inferences drawn from them are not amenable to formal statistical treatment or parametric generalizations.

Resource Optimization: It offers a distinct practical advantage in terms of saving time, money, and administrative energy.

Significance

Resource Efficiency: Offers distinct practical advantages in conserving time, money, and energy.

Conditional Reliability: Yields tolerably reliable results if investigators remain strictly impartial, unbiased, and experienced.

Applicability

Recommended For: Small inquiries, individual research projects, or studies requiring intensive investigation of a small universe.

Avoided For: Large-scale inquiries of major importance.

Example:

To study the economic conditions of an entire state, an investigator purposively selects a few specific towns and villages for intensive analysis, judging them to be a miniature cross-section of the whole.

Types of Non-Probability Sampling Design:

1. Convenience / Haphazard Sampling

Meaning: An unrestricted non-probability design where the researcher selects easily accessible individuals from the population at large without any rigid aim or defined purpose.

Key Features: * It does not require a structured or randomized selection framework.

Units are chosen dynamically based purely on their immediate, real-time availability.

Significance: * It offers significant practical advantages by conserving time, money, and administrative effort.

It allows for rapid, baseline data collection without complex preparatory tracking.

Applicability: * It is highly applicable for individual research projects or small-scale exploratory inquiries.

It is used when creating a complete population source list is physically or practically impossible.

Illustration: Selecting a group of trees in a vast forest completely at random or haphazardly to estimate their mean height because numbering every individual tree to draw a random sample is impossible.

2. Purposive / Judgement Sampling

Meaning: A sampling procedure where the researcher deliberately selects specific items from the universe because they judge them to be typical or representative of the whole.

Key Features:

The researcher's personal choice and expertise remain supreme during unit selection.

It carries a high risk of personal bias entering the selection process, which can potentially vitiate the inquiry.

Significance:

It yields tolerably reliable results if the investigator is strictly impartial, experienced, and exercises sound judgment.

It eliminates the need for expensive, expansive randomized setups when targeting specific data traits.

Applicability:

It is most appropriate when the target universe is relatively small.

It is ideal when a known, specific characteristic of a population needs to be studied intensively.

Illustration: If an investigator wants to study the economic conditions of an entire state, they purposively select a few specific towns and villages for deep analysis based on the expert judgment that these areas mirror the state's broader cross-section.

3. Quota Sampling

Meaning: A restricted non-probability design where field interviewers are assigned explicit category quotas to be filled from designated population strata.

Key Features:

While specific structural restrictions are set on the demographic strata, the actual final selection of the individual items is left entirely to the interviewer's discretion.

The resulting samples do not possess random traits and are not amenable to formal statistical treatment.

Significance:

It is exceptionally convenient, fast to coordinate, and highly cost-effective.

It ensures that distinct sub-groups within a population are visibly accounted for in the final count without randomized stratification logistics.

Applicability:

It is widely utilized in public opinion polls, commercial market research, and rapid field surveys where explicit demographic representation is required.

Illustration: A researcher sets a structural field quota requiring exactly 50 males, 50 females, 60 individuals aged 18–30, and 60 individuals aged 31–50; the field interviewers then approach anyone available on the street who matches those criteria until each target bracket is full.

4. Snowball Sampling

Meaning: A chain-referral non-probability mechanism where initial subjects are picked to recruit subsequent eligible participants from their own social or professional networks.

Key Features:

It operates via sequential recruitment waves, causing the total sample size to accumulate and grow progressively larger over time.

Selection relies heavily on the interconnected networks of existing participants rather than independent selection rules.

Significance:

It penetrates hard-to-reach populations where a formal source list or sampling frame does not exist.

It bridges critical participant trust barriers by utilizing personal introductions and peer-to-peer validation.

Applicability:

It is highly ideal for studying rare, hidden, or highly insulated target populations that are not easily mapped out or visible to the public.

Illustration: Conducting a study on a niche group (such as pioneer angel investors or specialized technology founders) by finding one initial participant who fits the criteria, interviewing them, and then asking them to refer other eligible peers from their professional network to participate next.

Sample Size:

Meaning: Sample size refers explicitly to the total number of items or observational units selected from the universal population to constitute a study's sample. Determining size is a major challenge for researchers; a sample size must neither be excessively large nor too small. A researcher must always strive to design an optimum sample. An optimum sample is one that simultaneously fulfils four fundamental criteria: efficiency, representativeness, reliability, and flexibility.

Determination of Sample Size

Sample size determination is the process of selecting the exact number of observations (n) to include in a statistical sample. It directly dictates the balance between statistical precision and operational resource constraints (time, budget, and manpower) in business research methods.

Before determining an appropriate sample size, a researcher must clearly define certain statistical and methodological parameters. These parameters ensure that the selected sample accurately represents the target population and produces reliable research findings.

1. Population Size (N)

Population size refers to the total number of individuals, units, or elements included in the target group under study. For example, it may represent the total number of customers, employees, investors, or credit card users of a bank.

When the population size is known and limited, it is treated as a finite population.

When the population is extremely large or unknown, it is considered an infinite population for statistical calculation purposes.

2. Margin of Error (E or e)

The margin of error, also called the precision level or sampling error, indicates the maximum acceptable difference between the sample result and the actual population value. It reflects the accuracy expected from the study findings.

Commonly used margins of error are:

±5% — acceptable for most management and social science studies

±3% — used when higher precision is required

±1% — used in highly sensitive or large-scale studies

A smaller margin of error increases the required sample size.

3. Confidence Level

Confidence level represents the probability that the sample findings truly reflect the characteristics of the population within the specified margin of error. It indicates the reliability of the research results.

The most commonly used confidence levels in MBA and business research are:

90% Confidence Level

95% Confidence Level

99% Confidence Level

Higher confidence levels require larger sample sizes.

4. Z-Score (Z)

The Z-score is the standardized statistical value corresponding to the selected confidence level under the normal distribution curve. It is used in sample size calculation formulas.

Common Z-values are:

5. Population Variability (σ or p)

Population variability refers to the degree of variation or heterogeneity within the population.

For continuous data, variability is represented using the standard deviation (σ).

For categorical or attribute-based data, variability is represented using the estimated population proportion (p).

When the actual proportion is unknown, researchers commonly assume:

$p=0.5$

This assumption provides the maximum possible variability, resulting in the largest and safest sample size for the study.

$p=0.5$

Importance of Determining Sample Size

Determining an appropriate sample size is one of the most critical steps in research design because it directly influences the accuracy, reliability, validity, and practicality of the study. An inadequately chosen sample may lead to misleading conclusions, while an excessively large sample may waste time and resources. The importance of sample size determination can be explained as follows:

1. Improves Accuracy and Reliability of Research Findings

A properly determined sample size increases the likelihood that the sample characteristics closely reflect the actual characteristics of the population. Larger and scientifically selected samples reduce random fluctuations and provide more consistent results.

For example, if a bank wants to study customer satisfaction, collecting responses from only 10 customers may produce unreliable findings. However, surveying an adequately calculated sample improves the credibility and dependability of the conclusions.

Thus, an appropriate sample size enhances:

Accuracy of estimates

Consistency of results

Reliability of research outcomes

2. Minimizes Sampling Error

Sampling error refers to the difference between the sample result and the true population parameter. A well-calculated sample size helps reduce this error and increases the precision of the study.

When the sample size increases:

The margin of error decreases

Estimates become more stable

Research findings become more representative of the population

For instance, election opinion polls with very small samples may produce inaccurate predictions, whereas scientifically calculated samples provide more dependable forecasts.

3. Ensures Better Representation of the Population

An adequate sample size improves the representativeness of the study by ensuring that different categories, opinions, and characteristics within the population are sufficiently covered.

A representative sample:

Reflects population diversity

Reduces bias

Enhances generalizability of results

For example, in employee satisfaction research, a proper sample should include employees from different departments, age groups, and experience levels. If the sample is too small, certain groups may be excluded, leading to biased conclusions.

4. Supports Valid Statistical Analysis and Hypothesis Testing

Statistical tools such as correlation, regression, t-tests, chi-square tests, ANOVA, and hypothesis testing require sufficient sample sizes for meaningful interpretation.

An inadequate sample size may:

Reduce statistical power

Increase the possibility of Type I and Type II errors

Produce unstable or insignificant results

A properly determined sample size ensures:

Better statistical validity

Higher confidence in hypothesis testing

More meaningful interpretation of relationships and trends

For example, in MBA research involving customer buying behavior, regression analysis performed on a very small sample may fail to detect actual relationships among variables.

5. Saves Time, Cost, and Research Resources

While larger samples generally improve accuracy, collecting unnecessary data increases:

Cost of data collection

Time required for fieldwork

Data processing complexity

Researcher workload

Sample size determination helps achieve an optimal balance between:

Accuracy and precision

Cost and feasibility

Time and resource utilization

Thus, researchers can avoid both:

Under-sampling (too few respondents leading to weak conclusions)

Over-sampling (excessively large samples causing resource wastage)

6. Enhances Decision-Making Quality

In management, business, healthcare, finance, and social science research, decisions are often based on research findings. A scientifically determined sample size improves the quality of managerial and policy decisions because the findings are more dependable and evidence-based.

For example:

Marketing decisions rely on customer surveys

HR decisions depend on employee feedback studies

Financial decisions use investor perception analysis

Accurate sample sizes ensure these decisions are supported by credible data.

7. Increases Research Credibility and Academic Acceptance

Research studies with scientifically justified sample sizes are considered more professional and academically acceptable. Universities, journals, and research institutions often evaluate whether the sample size is statistically adequate before accepting research findings.

Proper sample size determination:

Strengthens research methodology

Improves academic quality

Increases acceptance of findings

Enhances the credibility of the researcher

Conclusion

Sample size determination is essential for conducting effective and scientifically valid research. It ensures accurate representation of the population, reduces sampling error, supports reliable statistical analysis, and optimizes research resources. Therefore, selecting an appropriate sample size is a fundamental requirement for producing credible, meaningful, and generalizable research findings.

Basic Methods of Estimating Sample Size for Different Situations

Sample size estimation varies depending on the nature of the research problem, type of data, and population characteristics. Different statistical formulas are used for different research situations such as estimating proportions, means, finite populations, or unknown populations.

1. Sample Size Estimation for Infinite Population (Proportion Method)

This method is used when:

The population is very large or unknown

The study involves categorical responses such as Yes/No, Satisfied/Unsatisfied, etc.

Common in marketing and social science research

Formula

Where:

n = Required sample size

Z = Z-value corresponding to confidence level

p = Estimated population proportion

$q = 1 - p$

e = Margin of error

Illustration

A researcher wants to study customer satisfaction with:

95% confidence level

5% margin of error

Population proportion unknown

Since p is unknown:

$p = 0.5$

$Z = 1.96$

$e = 0.05$

Substituting:

Result

The required sample size is 385 respondents.

2. Sample Size Estimation for Finite Population (Cochran's Method)

Used when:

Population size is known and limited

Example: employees in a company, students in a college, bank customers, etc.

Formula

Where:

Illustration

A researcher wants to estimate the average monthly expenditure with:

Standard deviation (σ) = ₹500

Confidence level = 95%

Margin of error (e) = ₹50

For 95% confidence level:

$$Z=1.96$$

Substituting into the formula:

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Thus, the initial sample size for an infinite population is:

$$n_0 = 385$$

Assume the total population size is:

$$N = 2000$$

Using the finite population correction formula:

Final Answer

$$\text{Initial sample size } (n_0) = 385$$

$$\text{Adjusted sample size for finite population } (n) = 323 \text{ respondents}$$

Thus, when the population size is limited to 2000, only 323 respondents are required instead of 385.

Rule of Thumb Method

This is a simplified practical approach used when statistical information is unavailable.

Common rules include:

Minimum 30 samples for basic statistical analysis

5–10 respondents per questionnaire item

100–200 samples for MBA and social science studies

Regression analysis:

Minimum $50 + 8m$

m = number of independent variables

Illustration

If a study has:

5 independent variables

Then:

$$n = 50 + 8(5) = 50 + 40 = 90$$

Result

Minimum sample size required = 90 respondents.

Summary of Basic Sample Size Estimation Methods

Sampling vs. non-sampling Errors

Nature and Definition

Sampling Errors: These are random variations in the sample estimates around the true population parameters. They occur purely by blind chance, meaning they are equally likely to be in either direction and are of a compensatory type. The expected value of these random errors is equal to zero.

Systematic Bias (Non-Sampling Errors): This results from actual errors in the sampling procedures or execution. Unlike random errors, a systematic bias constantly skews the data in a particular direction.

Impact of Sample Size

Sampling Errors: These errors decrease in magnitude as the size of the sample increases. Increasing the sample size is a direct way to improve the precision of your sampling plan.

Systematic Bias (Non-Sampling Errors): This bias cannot be reduced or eliminated by increasing the sample size. In fact, if a systematic bias is present, the error will get larger and larger as the number of observations increases.

Primary Causes

Sampling Errors: Caused by random chance and the inherent nature of studying only a part of a population rather than the whole. They are also influenced by the population itself; for instance, sampling errors are of a smaller magnitude if the population is highly homogeneous.

Systematic Bias (Non-Sampling Errors): Usually the result of operational and procedural factors, including:

Inappropriate sampling frame: Using a biased list or source list that does not accurately represent the universe.

Defective measuring devices: Using a constantly flawed physical device, a biased questionnaire, or an interviewer who influences responses.

Non-respondents: Failing to establish contact or receive a response from individuals initially included in the sample, especially when the reason they aren't responding correlates with what you are trying to estimate.

Indeterminacy principle: Individuals acting differently simply because they know they are being observed.

Natural bias in reporting data: The tendency of respondents to intentionally understate or overstate figures (e.g., understating income for tax surveys or overstating it for social status).

Measurement and Control

Sampling Errors: Can be mathematically measured for a given sample design and size. The effective way to control them without ballooning data collection costs is to choose a better sampling design that inherently yields a smaller sampling error.

Systematic Bias (Non-Sampling Errors): Cannot be mathematically measured in the same way. At best, the specific causes responsible for these errors must be actively detected and corrected by the researcher.

Summary

Steps and Critical Considerations for Developing an Optimal Sample Design

Developing a reliable and appropriate sample design requires a researcher to pay close attention to several critical steps, constraints, and foundational characteristics.

Here are the key points that must be considered:

1. Key Steps and Parameters in Sample Design

Type of Universe: The first step is to clearly define the set of objects to be studied, which is technically called the Universe. The universe can be finite (where the number of items is certain, like the population of a city) or infinite (where the number of items is unlimited, like throwing a dice or the stars in the sky).

Sampling Unit: A decision must be made regarding the specific unit to be selected for the sample. This can be a geographical unit (state, district, village), a construction unit (house, flat), a social unit (family, club, school), or an individual respondent.

Source List (Sampling Frame): The researcher must identify or prepare a source list from which the sample will be drawn. This list contains the names of all items in a finite universe and must be comprehensive, correct, reliable, appropriate, and as representative of the population as possible.

Parameters of Interest: The specific population parameters to be estimated must be determined. For example, the researcher needs to know if the study is focusing on a specific proportion of the population, an average measure, or specific estimations for important sub-groups.

Sampling Procedure: The final decision involves choosing the actual technique or procedure to select the sample items. The researcher should aim to select a design that minimizes sampling error for a given sample size and cost.

2. Determining the Size of the Sample

Deciding on an optimum sample size—one that fulfills the requirements of efficiency, representativeness, reliability, and flexibility—is a major problem that requires balancing multiple factors:

Desired Precision and Confidence Level: The researcher must decide on the acceptable confidence level and precision required for the final estimates.

Population Variance: If the population has a larger variance (highly heterogeneous), a bigger sample size is typically needed to maintain accuracy.

Population Size: The overall size of the universe inherently limits and dictates the boundaries of the sample size.

3. Managing Practical Constraints and Costs

Budgetary Constraints: Cost considerations heavily dictate both the size and the type of sample a researcher can draw. If funds are highly restricted, budgetary limits can even necessitate the use of a non-probability sample instead of a probability sample.

Cost of Incorrect Inference: The researcher must weigh two types of costs: the cost of collecting data versus the financial or analytical cost of making an incorrect inference based on the data.

4. Controlling Errors and Biases

A good sample design must balance and control two causes of incorrect inferences:

Systematic Bias (Non-Sampling Error): This results from procedural errors (like an inappropriate sampling frame, a defective questionnaire, non-respondents, or natural reporting biases) and cannot be fixed by simply increasing the sample size. The design must actively minimize these factors.

Sampling Error: These are the random variations that naturally occur because you are studying a sample instead of a census. The design should minimize this error, keeping in mind that while a larger sample size reduces sampling error, it simultaneously increases data collection costs and can enhance systematic bias if not carefully managed.

Summary Checklist of a Good Sample Design

Ultimately, the final sample design should meet the following five baseline characteristics:

It must result in a truly representative sample.

It must result in a small sampling error.

It must be viable and realistic in the context of available funds.

It must allow systematic bias to be controlled effectively.

It must allow the results to be generalized to the universe with a reasonable level of confidence.